TalentLink serves as a Learning Management System (LMS) that personalizes the academic experience for students and faculty.





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Session 2024-2028

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TalentLink:
A Learning Management System





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May, 2026

This thesis is submitted in partial fulfillment of the Requirement of the Bachelors of studies in Software Engineering,
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DEDICATION

With deepest gratitude, I honor my parents, whose love, wisdom and unwavering support have been the driving force behind my academic journey. Their guidance, encouragement, and sacrifices have shaped me into the individual I am today. I am forever thankful for their presence in my life.

DECLARATION

I hereby declare that this thesis is my own work and that, to the best of my knowledge and belief, it contains no material previously published or written by another person, nor material which to a substantial extent has been accepted for the award of any other degree or diploma of any university or other institution, except where due acknowledgment is made in the text. I also declare that the intellectual content of this thesis is the product of my own work, except to the extent that assistance from others in the project's design and conception or in style, presentation, and linguistic expression is acknowledged.

The work presented in this thesis was carried out by me under the supervision of Prof. Faisal Hameed, Department of Computer Science, Government Islamia Graduate College, Kasur. These conclusions are my own research after numerous discussions with my supervisor. I have not presented any part of this work for any other degree. Furthermore, I acknowledge that any assistance received in preparing this thesis and the sources from which such assistance was derived are acknowledged within the thesis itself. I certify that to the best of my knowledge above statement is correct.

Laiba Fatima

ACKNOWLEDGEMENTS

I am eternally grateful to Almighty ALLAH for bestowing upon me the strength, potential, and capability to accomplish my research work. I am proud to be an apostle of the world's greatest personality, Holy Prophet Muhammad (PBUH), who is mankind's only full and absolute personality.

I am grateful to Prof. Muhammad Shabeer, Head of the Department of Computer Science, for his knowledge and teaching experience. I'd want to thank him for his thoughtful and positive attitude. I am also thankful to Prof. Faisal Hameed for their guidance and knowledge throughout the duration of the coursework and research work. This research, which is neither simple nor straightforward, has provided me with the opportunity to interact with many academically minded individuals and gain valuable life experience. I would want to convey my heartfelt gratitude to my renowned supervisor, Prof. Faisal, who helped me in various ways to complete my task. It would have been difficult to finish without his involvement and aid. My heartfelt gratitude goes to my family. This job will never be finished without the prayers, support, and courage of my esteemed parents and close brother. May Allah shower him with His innumerable blessings.

CERTIFICATION

This is to certify that the thesis titled 'TalentLink:
A Learning Management System' authored by Laiba Fatima, has been completed in accordance with the requirements for the BS in Computer Science.

I hereby confirm the following:

- The thesis represents the original work of the author.
- Any assistance received in the preparation of this thesis has been duty acknowledge.
- All sources used in this thesis have been properly cited and referenced.
- No part of this thesis has been previously submitted for any other academic qualification.
- The content of this thesis does not infringe upon any proprietary rights, including copyright or trademark.

This certification is issued upon the successful completion and examination of the thesis, meeting the academic standards set forth by the Computer Science Department.

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1 Contents

1 Introduction 1

1.1 Purpose 1

1.2 Document Conventions 1

1.3 Intended Audience 1

1.4 Definitions, Acronyms, and Abbreviations 1

1.5 Scope 2

1.6 References 2

2 Overall Description 2

2.1 Product Perspective 3

2.1.1 User Interface 3

2.1.2 System Interface 3

2.2 Product Functions 3

2.3 User Characteristics 4

2.4 Operating Environment 4

2.5 Design and Implementation Constraints 4

2.6 Design Methodologies 5

2.7 Assumption and Dependencies 5

3 Specific Requirements 6

3.1 Functional Requirements 6

3.2 External Interface Requirements 6

3.2.1 User Interface 6

3.2.2 Hardware Interface 6

3.2.3 Software Interface 7

3.2.4 Communication Interface 7

3.3 System Features 7

3.3.1 User Authentication System 7

3.3.2 Course Content Management 7

3.3.3 Assignment Submission Tracking 7

3.3.4 Grading and Feedback Portal 8

3.3.5 Notification and alert System 8

3.3.6 Online quiz and Assessment Module 8
3.3.7 Examination and Date Sheet Management 8
3.3.8 Interactive To Do List & Activity Tracker 9
3.3.9 Fee Management & Billing System 9
3.3.10 Attendance Tracking System 9
3.3.11 User Profile and Account Setting 9
3.3.12 Digital Grade Book & Performance Analytics 10
3.3.13 Academic Course History 10
3.3.14 Contact Us & Support System 11
3.4 Non Functional Requirements 11
3.4.1 Performance Requirements 11
3.4.2 Design Constraints 12
3.4.3 Software System Attributes 12
3.5 Other Requirements 13
3.5.1 Documentation and User Manuals 13
3.5.2 Legal and Copyright Compliance 13
3.5.3 Operational Maintenance 13
3.5.4 Localization and Language 13
4 System Models 13
4.1 GENERALIZED DIAGRAM 14
4.2 Login Diagrams 16
4.2.1 USE-CASE DIAGRAM FOR LOGIN 16
4.2.2 Login Sequence Diagram 17
4.2.3 Login Activity Diagram 18
4.2.4 User Account State Diagram 19
4.3 Assignment Submission Diagrams 20
4.3.1 Submit Assignment – Use Case 20
4.3.2 Submit Assignment – Sequence diagram 21
4.3.3 Submit Assignment__Activity Diagram 22
4.3.4 Submit Assignment State Diagram 23
4.4 Send Notification Diagrams 24
4.4.1 Use Case – Send Notification 24
4.4.2 Send Notifications – Sequence Diagram 26
4.4.3 Send Notifications – Activity Diagram 27

4.4.4 Notification State Diagram 28
4.5 Data Flow Diagrams 29
4.5.1 Level 0 __ Context Diagram DFD 29
4.5.2 Level 1 __ System Level DFD 30
4.5.3 Level 3 __Detailed Level DFD 31
4.6 Class Diagram 33
4.7 ERD 34
4.8 Gantt Chart 35
4.9 Test Cases: 36
4.9.1 User Login Test Case 36
4.9.2 Submit Assignment Test Case 37
4.9.3 Send Notification Test Case 38

1 Introduction

1.1 Purpose

The primary purpose of this document is to define the software requirements for the **TalentLink** platform. **TalentLink** is developed to bridge the gap between skilled individuals and professional opportunities by providing a centralized digital ecosystem. It aims to simplify the process of talent discovery through an automated and user-friendly interface. This SRS ensures that all stakeholders, including developers and supervisors, have a clear understanding of the system's functionality and goals.

1.2 Document Conventions

This document follows the standard IEEE format for Software Requirements Specifications (SRS). To maintain professional clarity and consistency, the following conventions are applied:

- **Heading Styles:** All primary and secondary headings are formatted in **Blue** color to enhance the document's scannability and distinguish them from the body text.
- **Typography:** The standard font used for body text is Times New Roman, size 12, while headings use a larger font size for better hierarchy.
- **Emphasis:** Important terms, including the project name **TalentLink**, are highlighted in bold to draw attention to key concepts.

1.3 Intended Audience

This document is intended for various stakeholders involved in the design and development of **TalentLink**. The primary audience includes:

- **Project Supervisors:** To review the document and ensure it meets the academic standards of **Govt. Islamia College**.
- **Software Developers:** To understand the functional and technical requirements needed for the implementation of the platform.
- **Quality Assurance (QA) Teams:** To use the requirements defined here as a basis for creating test cases and ensuring software reliability.
- **End Users:** To get an overview of the system's capabilities and how it connects talent with opportunities.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
LMS	Learning Management System
UI/UX	User Interface and User Experience – The visual design and how easily a user can navigate through the platform.
Database	A structured collection of data, such as user profiles and skill sets, stored and accessed electronically.

Authentication	The process of verifying the identity of a user attempting to access the TalentLink system.
Functional requirements	The specific services or features that the TalentLink system must provide to the users.

1.5 Scope

The scope of **TalentLink** is to develop a centralized web-based platform that facilitates the connection between skilled individuals and professional opportunities. The system is designed to streamline the process of talent discovery and recruitment. Key features and goals of the project include:

- **User Profile Management:** Allowing users to create and maintain digital profiles that highlight their skills, education, and experience.
- **Skill-Based Search:** A powerful filtering system that enables recruiters or organizations to find the right talent based on specific keywords or categories.
- **Secure Authentication:** Implementing a robust login and registration system to ensure data privacy and secure access for all users.
- **Interactive Dashboard:** A user-friendly interface for both admins and users to manage activities and track professional connections.
- **Communication Bridge:** Providing a platform where initial professional contact can be established efficiently.

1.6 References

- **IEEE Std 830-1998:** IEEE Recommended Practice for Software Requirements Specifications.
- **Software Engineering: A Practitioner's Approach:** By Roger S. Pressman, for understanding the software development life cycle and requirement gathering.
- **LMS Functional Standards:** General documentation regarding Learning Management System features like user roles, course management, and resource allocation.
- **Database Management System (DBMS) Concepts:** For referencing data storage, student records, and relational database structures used in the LMS.

2 Overall Description

This provides an overview of the system's functionality, its relationship to other systems, and the constraints under which it must operate.

- **System Context:** The LMS is a standalone web-based platform designed to digitalize and automate existing manual educational and administrative processes.
- **Centralized Architecture:** The system utilizes a centralized **MySQL database** to securely store and manage all academic data, ensuring that information is accessible to authorized users in real-time.
- **Operational Goal:** The primary objective is to eliminate paper-based documentation and facilitate seamless resource sharing and communication between faculty and students.

- **Mode of Operation:** The application is designed to function in a web-based environment, making it accessible via both local networks and the internet.

2.1 Product Perspective

The Learning Management System (LMS) is a new, self-contained, and standalone software product designed to replace manual academic coordination. It does not depend on any pre-existing software but is built to integrate with modern web technologies and database servers to provide a unified educational platform.

2.1.1 User Interface

The system provides a graphical user interface (GUI) that is:

- **Intuitive & Responsive:** Accessible via desktops, laptops, and tablets.
- **Dashboard-Driven:** Users are presented with a personalized dashboard upon login to access courses, notifications, and profiles.
- **Navigation-Friendly:** A consistent sidebar and menu system ensure that users can switch between modules easily.

2.1.2 System Interface

To function effectively, the LMS interacts with the following external components:

- **Database Interface:**

The application communicates with a **MySQL database** for real-time data storage, including user credentials and academic records.

- **File System Interface:**

The system interacts with the server's local storage to manage, save, and retrieve uploaded documents like PDF notes and assignment submissions.

- **Communication Interface:**

The system uses the HTTP/HTTPS protocols to facilitate secure communication between the user's browser and the web server.

2.2 Product Functions

The Learning Management System (LMS) provides a wide range of functions to facilitate the educational process. The major functions of the system are categorized as follows:

- **User Authentication and Authorization:** Secure login and logout mechanisms for students, teachers, and administrators to ensure that users can only access data relevant to their roles.
- **Profile Management:** Allows users to view and update their personal information, change passwords, and manage their academic profiles.
- **Course Management:** Enables teachers to create, update, and organize course materials, including lecture notes, syllabus, and reading resources.
- **Student Enrollment:** Automates the process of enrolling students into specific courses and maintaining an active list of participants for each class.
- **Assignment Submission and Grading:** Provides a platform for students to upload assignments digitally and allows teachers to grade them and provide feedback within the system.
- **Content Distribution:** Acts as a central repository where educational content

(PDFs, PPTs, Images) can be shared by faculty and downloaded by students.

- **Progress Tracking and Reports:** Generates reports on student performance, attendance, and overall course progress for both students and instructors to review.

2.3 User Characteristics

The Learning Management System (LMS) is designed to be used by three primary groups of users. Each group has different responsibilities and requires a specific level of technical proficiency:

- **Administrator:** The Administrator is the most powerful user with full access to the system. Their responsibilities include managing user accounts (creating and deleting profiles), monitoring system security, and maintaining the database. They are expected to have a high level of technical expertise and an understanding of system configurations.
- **Faculty Members (Teachers):** Teachers use the system to deliver educational content. Their main tasks include uploading lecture notes, creating assignments, and grading student work. They should have basic computer literacy, such as file management and experience with web browsers, to effectively manage their virtual classrooms.
- **Students:** Students are the primary consumers of the system. They use the platform to download resources, submit assignments, and view their academic progress. They are expected to have basic digital skills, including the ability to navigate a web interface and upload documents in various formats (e.g., PDF, DOCX).

2.4 Operating Environment

This section describes the hardware and software environment required for the Learning Management System to function effectively:

- **Client Side (User's Device):** The system is platform-independent and can run on any device (Desktop, Laptop, or Tablet) that supports a modern web browser.
- **Browsers:** Google Chrome, Mozilla Firefox, Microsoft Edge, or Safari.
- **Operating System:** Windows, macOS, Linux, Android, or iOS.
- **Server Side:** The backend of the application will be hosted on a server environment capable of processing requests and managing the database.
- **Web Server:** Apache or Nginx.
- **Database Server:** MySQL Server 8.0 or higher.
- **Runtime Environment:** The server must support the languages used for development (e.g., PHP, Python, or Node.js).
- **Network Requirements:** A stable internet or Intranet (Local Area Network) connection is required for users to access the portal and for the client-server communication to take place.

2.5 Design and Implementation Constraints

These are the limitations and requirements imposed on the design and implementation of the system:

- **Security Constraint:** All user passwords must be stored in the database using

strong encryption (e.g., hashing) to prevent unauthorized access to sensitive user data.

- **Database Constraint:** The system must use the **MySQL** relational database management system for all data storage and retrieval tasks to ensure data integrity.
- **Browser Compatibility:** The web interface must be compatible with modern browsers like Google Chrome, Mozilla Firefox, and Safari, and it must conform to HTML5 and CSS3 standards.
- **Platform Independence:** The application should be developed as a web-based system so that it can be accessed from any operating system (Windows, Linux, macOS) without needing a local installation.
- **Scalability:** The system design must allow for future expansion, such as adding more user roles or increasing the number of supported courses without requiring a complete redesign.

2.6 Design Methodologies

The TalentLink Learning Management System will be developed using the **Waterfall Model**. This model was selected because all system requirements are clearly defined at the beginning of the project, and the scope is not expected to change during development. The Waterfall Model provides a structured and sequential approach which is best suited for academic projects like TalentLink.

The development will follow these sequential phases:

- **Phase 1 — Requirements Gathering:** All functional and non-functional requirements of the TalentLink LMS have been collected, analyzed, and documented in this SRS document.
- **Phase 2 — System Design:** The overall system architecture, database schema (MySQL), and user interface layouts will be designed based on the defined requirements.
- **Phase 3 — Implementation:** All 14 system modules will be developed and coded according to the design specifications using HTML5, CSS3, JavaScript, Python and PHP.
- **Phase 4 — Testing:** Each module will be tested individually (Unit Testing) and then as a complete system (Integration Testing).
- **Phase 5 — Deployment:** The completed system will be deployed on a web server and made accessible to Admin, Teachers, and Students.
- **Phase 6 — Maintenance:** After deployment, the system will be monitored for bugs, performance issues, and future enhancements.

2.7 Assumption and Dependencies

This section outlines the factors that are assumed to be true for the project's success and the external elements that the system depends on:

- **Assumptions:**
  - **User Literacy:** It is assumed that all users (Administrators, Teachers, and Students) possess basic computer skills and are familiar with navigating web-based interfaces.
  - **Device Availability:** It is assumed that users have access to a compatible device (PC, Laptop, or Smartphone) to access the LMS portal.

- **Data Accuracy:** It is assumed that the information provided by users during registration and profile setup is accurate and valid.

- **Dependencies:**
  - **Internet Connectivity:** The system's performance and accessibility are highly dependent on a stable internet or local area network connection.
  - **Server Uptime:** The availability of the LMS depends on the continuous operation of the web server and the **MySQL** database server.
  - **Browser Support:** The system relies on the user having a modern, updated web browser that supports HTML5 and JavaScript.

3 Specific Requirements

This chapter details the specific requirements for the Learning Management System. It includes functional requirements, external interface requirements, and system attributes.

3.1 Functional Requirements

The functional requirements of TalentLink LMS are organized into 14 system modules, each detailed in Section 3.3. The following provides a high-level summary of the core system capabilities:

- **User Management:** The system shall allow the administrator to create, update, and delete user profiles for teachers and students.
- **Course Management:** Teachers shall be able to upload educational content, including documents and links, and organize them by course.
- **Assignment Handling:** The system shall enable students to upload assignments and allow teachers to grade them and provide feedback.
- **Dashboard Access:** Users shall be provided with a personalized dashboard showing their enrolled courses, pending tasks, and recent grades.

3.2 External Interface Requirements

This section provides a detailed description of the interfaces between the Learning Management System (LMS) and the external entities, including users, hardware, and software environments.

3.2.1 User Interface

- **Design Philosophy:** The system shall feature a clean, minimalist web interface designed with a "Mobile-First" approach to ensure usability across all screen sizes.
- **Consistency:** A unified color scheme and typography shall be maintained throughout the portal to provide a professional user experience.
- **Navigation:** A persistent sidebar or navigation menu shall be provided for quick access to the Dashboard, Courses, Assignments, and Profile settings.
- **Feedback Mechanism:** The UI shall provide immediate visual feedback (e.g., loading spinners, success toasts, or error messages) for every user action.

3.2.2 Hardware Interface

- **Processing Power:** The system shall be optimized to run on devices with a minimum of 2GB RAM and a dual-core processor.
- **Display:** The interface must automatically adjust its layout for various resolutions,

supporting a minimum width of 320px (smartphones) and up to 1920px (FHD Monitors).

- **Peripherals:** The system shall support standard input devices, including keyboards, pointing devices (mouse/trackpad), and touchscreens for mobile interaction.

3.2.3 Software Interface

- **Operating Systems:** The system shall be platform-agnostic, supporting Windows, macOS, Linux, Android, and iOS through a standard web browser.
- **Database Management:** The application shall interface with **MySQL 8.0+** using secure connection strings for all CRUD (Create, Read, Update, Delete) operations.
- **Web Server:** The system shall be configured to run on an **Nginx** or Apache environment, utilizing PHP or Node.js as the runtime engine.
- **Document Rendering:** The interface shall support the viewing of PDF and common image formats natively within the browser using standard web APIs.

3.2.4 Communication Interface

- **Network Protocols:** All data exchange between the client and the server shall occur over **HTTPS** to ensure end-to-end encryption.
- **Email Integration:** The system shall utilize SMTP (Simple Mail Transfer Protocol) for sending automated notifications, such as password resets and assignment reminders.
- **Data Format:** The system shall use **JSON** (JavaScript Object Notation) for efficient data exchange between the frontend and the backend API.

3.3 System Features

3.3.1 User Authentication System

- **Description:**
  - This feature ensures that only authorized users can access the system resources.
- **Functional Requirements:**
  - The system shall provide a login portal for Students, Teachers, and Administrators.
  - The system must validate user credentials against the **MySQL** database.
  - The system shall support session management to automatically log out users after a period of inactivity.

3.3.2 Course Content Management

- **Description:**
  - This is the core feature for teachers to organize and distribute academic materials.
- **Functional Requirements:**
  - Teachers shall be able to create course categories and upload files (PDF, PPT, DOCX).
  - Students shall have the ability to view and download these materials based on their enrollment status.
  - The system shall allow the Admin to monitor and manage all uploaded content.

3.3.3 Assignment Submission Tracking

- **Description:**

- Facilitates the digital collection of student work and tracks deadlines.

- **Functional Requirements:**
  - The system shall allow teachers to set assignment titles, descriptions, and "Due Dates."
  - Students shall be able to upload their work as digital files before the deadline.
  - The system must flag any submissions made after the specified deadline as "Late."

3.3.4 Grading and Feedback Portal

- **Description:**
  - Provides a centralized location for evaluation and communication between faculty and students.
- **Functional Requirements:**
  - Teachers shall be able to view all student submissions for a specific assignment in one dashboard.
  - The system shall allow teachers to assign marks and provide written feedback.
  - The system shall automatically update the student's grade book upon the release of marks.

3.3.5 Notification and alert System

- **Description:**
  - Keeps users updated on important academic activities.
- **Functional Requirements:**
  - The system shall generate dashboard notifications for new course materials or assignments.
  - The system shall notify students of upcoming deadlines.
  - The system shall alert students once their assignments have been graded.

3.3.6 Online quiz and Assessment Module

- **Description:**
  - This feature allows teachers to conduct digital assessments to evaluate student progress in real-time.
- **Functional Requirements:**
  - The system shall allow teachers to create multiple-choice questions (MCQs) with a specific time limit.
  - The system must automatically shuffle questions to prevent cheating.
  - The system shall calculate and display the score immediately after the student submits the quiz.
  - The system must ensure that a student can only attempt a quiz once unless reset by the teacher.

3.3.7 Examination and Date Sheet Management

- **Description:**
  - Provides a centralized schedule for all upcoming mid-term and final examinations.
- **Functional Requirements:**
  - The Admin shall have the authority to upload and update the official Date Sheet for all departments.
  - The system shall provide a dedicated "Exam Portal" where students can view

their specific exam schedule based on their enrolled courses.
    - The system shall allow students to download the Date Sheet in PDF format for offline reference.
    - The system should notify users 24 hours before any scheduled exam.

3.3.8 Interactive To Do List & Activity Tracker

- **Description:**
    - This feature provides users with a personal task management tool to organize and track their upcoming academic responsibilities.
- **Functional Requirements:**
    - **Auto-Population:** The system shall automatically add "Upcoming Activities" to the To-Do list, such as assignment deadlines, quiz dates, and exam schedules.
    - **Manual Task Entry:** Users shall be able to manually add personal tasks (e.g., "Read Chapter 4" or "Meeting with Supervisor") with a specific priority level.
    - **Task Status:** Users must be able to mark tasks as "Completed," "In Progress," or "Pending."
    - **Reminder Alerts:** The system shall highlight tasks that are due within the next 24 hours to ensure timely completion.
    - **Deletion & Editing:** Users shall have the authority to edit or remove manually created tasks at any time.

3.3.9 Fee Management & Billing System

- **Description:**
    - This feature enables the administration to manage student financial records and allows students to track their payment status.
- **Functional Requirements:**
    - **Fee Status Tracking:** The system shall provide a dashboard for students to view their current balance, paid fees, and any outstanding dues.
    - **Voucher Generation:** The system shall allow students to download or print their fee challans/vouchers in PDF format.
    - **Payment History:** The system must maintain a digital record of all past transactions for each student.
    - **Admin Overlook:** The Administrator shall have the authority to update fee records, apply fines for late payments, or grant scholarships/discounts.
    - **Automated Alerts:** The system shall notify students via dashboard alerts regarding upcoming fee deadlines.

3.3.10 Attendance Tracking System

- **Description:**
    - A digital register to monitor the presence of students in different courses.
- **Functional Requirements:**
    - Teachers shall be able to mark daily attendance for their respective classes.
    - The system shall calculate the percentage of attendance for each student automatically.
    - Students must be able to view their own attendance record to ensure they meet the university's criteria.

3.3.11 User Profile and Account Setting

- **Description:**

- This feature allows users to manage their personal information and customize their account settings within the LMS.

- **Functional Requirements:**
  - **Profile Customization:** Users shall be able to upload a profile picture and update personal details such as contact number, email, and bio.
  - **Academic Snapshot:** The student profile shall display a summary of their current semester, GPA (if applicable), and a list of enrolled courses.
  - **Security Settings:** The system shall allow users to change their account password for security purposes.
  - **Role-Specific Info:** For teachers, the profile shall include their department, qualification, and the courses they are currently instructing.
  - **Activity Logs:** Users shall be able to view their last login time and recent activity for account monitoring.

3.3.12 Digital Grade Book & Performance Analytics

- **Description:**
  - This feature provides a comprehensive and automated record of all marks obtained by a student across various assessments.
- **Functional Requirements:**
  - **Consolidated View:** The system shall maintain a master grade book for each student, showing marks for Assignments, Quizzes, Mid-terms, and Finals in one table.
  - **Weighted Calculations:** The system shall automatically calculate the final grade based on pre-defined weightages (e.g., Assignments 20%, Quizzes 20%, Exams 60%).
  - **GPA/CGPA Calculation:** The system must automatically compute the Semester GPA and cumulative CGPA based on the marks and credit hours.
  - **Grade Privacy:** Students shall only be able to view their own grades, while teachers can view the entire class list.
  - **Export Options:** The system shall allow teachers to export the grade book in CSV or Excel format for official university records.
  - **Visual Analytics:** The system should provide a graphical representation (e.g., a bar chart or line graph) to show the student's performance trend over the semester.

3.3.13 Academic Course History

- **Description:**
  - This feature provides a complete archive of all the subjects a student has previously completed within the institution.
- **Functional Requirements:**
  - **Course Archive:** The system shall maintain a permanent record of all "Passed" or "Completed" courses from previous semesters.
  - **Grade Display:** For every studied course, the system shall display the final grade, marks obtained, and the specific semester/year it was completed.
  - **Instructor Reference:** The system shall show the name of the instructor who taught each of the previously studied courses.
  - **Credit Hour Audit:** The system shall provide a sum of all successfully completed credit hours against the total degree requirements.

- **Search & Filter:** Students shall be able to filter their studied courses by semester or by category (e.g., Core Courses, Electives, General Education).

3.3.14 Contact Us & Support System

- **Description:**
  - This feature provides a formal channel for users to seek assistance or report technical issues to the system administrators.
- **Functional Requirements:**
  - **Inquiry Form:** The system shall provide a "Contact Us" form where users can submit their name, email, subject, and a detailed message.
  - **Automated Ticketing:** Upon submission, the system shall generate a unique reference ID for the user's inquiry for tracking purposes.
  - **Department Routing:** Users shall be able to select the relevant department (e.g., Accounts, IT Support, Academic Office) to ensure the query reaches the right person.
  - **Admin Dashboard:** Administrators shall have a dedicated portal to view, manage, and respond to all user inquiries.
  - **Location & Info:** The page shall display the institution's official contact number, physical address (Govt. Islamia College), and email address for direct communication.

3.4 Non Functional Requirements

3.4.1 Performance Requirements

This section specifies the performance characteristics that the Learning Management System must exhibit to ensure a smooth and reliable user experience.

3.4.1.1 Response Time

The system shall respond to user requests (such as page loads or form submissions) within **2 to 3 seconds** under normal network conditions.

3.4.1.2 Capacity & Scalability

The system must be capable of handling at least **500 concurrent users** without any degradation in performance or system crashes.

3.4.1.3 Database Efficiency

Search queries and data retrieval from the **MySQL** database should be optimized to execute in less than **1 second** to ensure real-time access to records.

3.4.1.4 Resource Utilization

The application shall be optimized to run efficiently on standard web browsers without consuming excessive CPU or RAM on the client-side device.

3.4.1.5 Throughput

The system must support the simultaneous upload of multiple files (Assignments/Notes) by different users without bottlenecking the server bandwidth.

3.4.2 Design Constraints

This section describes the constraints imposed by the environment, hardware, and regulatory standards that limit the developer's options during the design phase.

3.4.2.1 Platform and Connectivity constraints

The Learning Management System (LMS) is strictly a web-based application. It must be designed to be accessible via standard HTTP/HTTPS protocols. The system's functionality is dependent on an active internet or intranet connection, and it must maintain a consistent user experience across different network speeds.

3.4.2.2 Technology stack Constraints

The development is constrained to a specific technology stack to ensure compatibility with the college's existing infrastructure. This includes using **HTML5, CSS3, and JavaScript** for the front-end, and a server-side language (such as **PHP**) for back-end logic. No proprietary or third-party frameworks should be used that require expensive licensing.

3.4.2.3 Database and Data Integrity constraints

The system must utilize the **MySQL** relational database management system (RDBMS). All data schemas must follow normalization rules to prevent data redundancy. The design is constrained to handle relational data mapping to ensure that student, course, and fee records remain synchronized at all times.

3.4.2.4 User Interface and Accessibility Constraints

The UI design must follow a **Responsive Web Design (RWD)** approach. It is constrained to be user-friendly for both technical and non-technical users (students and faculty). The layout must adapt to various screen resolutions (Mobile, Tablet, and Desktop) without losing critical navigational elements or functionality.

3.4.2.5 Security and Privacy Constraints

To comply with data protection standards, the system is prohibited from storing sensitive information like passwords in plain text. All authentication mechanisms must implement **encryption** or hashing. Additionally, the system must follow a **Role-Based Access Control (RBAC)** model, ensuring that users can only access data relevant to their specific role (e.g., Student, Teacher, or Admin).

3.4.3 Software System Attributes

In addition to performance, the following attributes are critical for the LMS:

3.4.3.1 Reliability

The system must ensure high data integrity, meaning that once an assignment or fee record is saved, it should not be lost or corrupted due to system errors.

3.4.3.2 Availability

The system shall maintain an uptime of **99.9%**, being accessible 24/7 except during brief, pre-announced maintenance windows.

3.4.3.3 Security

All user data and passwords must be protected using **encryption** (e.g., SHA-256 hashing). Access to administrative features must be restricted based on user roles.

3.4.3.4 Maintainability

The system shall be developed using a modular architecture and well-documented source code to ensure that future updates, bug fixes, and feature enhancements can be implemented with minimal effort and without affecting existing core functionalities.

3.4.3.5 Portability

Since the system is web-based, it must be fully functional across all major browsers, including Chrome, Firefox, Safari, and Edge.

3.5 Other Requirements

3.5.1 Documentation and User Manuals

The system shall be accompanied by a comprehensive User Manual that provides step-by-step instructions for students, teachers, and administrators on how to navigate and utilize the portal's features.

3.5.2 Legal and Copyright Compliance

The system must respect intellectual property rights. Any educational material, books, or resources uploaded to the Library module must comply with copyright laws and institutional policies.

3.5.3 Operational Maintenance

The system should include an automated backup mechanism that triggers every 24 hours to ensure that student records, grade books, and fee data are never lost in case of a server failure.

3.5.4 Localization and Language

While the primary interface of the LMS will be in English, the system should support Urdu text input where necessary (e.g., in student names or specific institutional notices) to cater to the local academic environment.

4 System Models

This section presents the visual models of the TalentLink Learning Management System. These diagrams illustrate the system's actors, database structure, and data flow to provide a clear understanding of the system's architecture and functionality.

4.1 GENERALIZED DIAGRAM

Description:

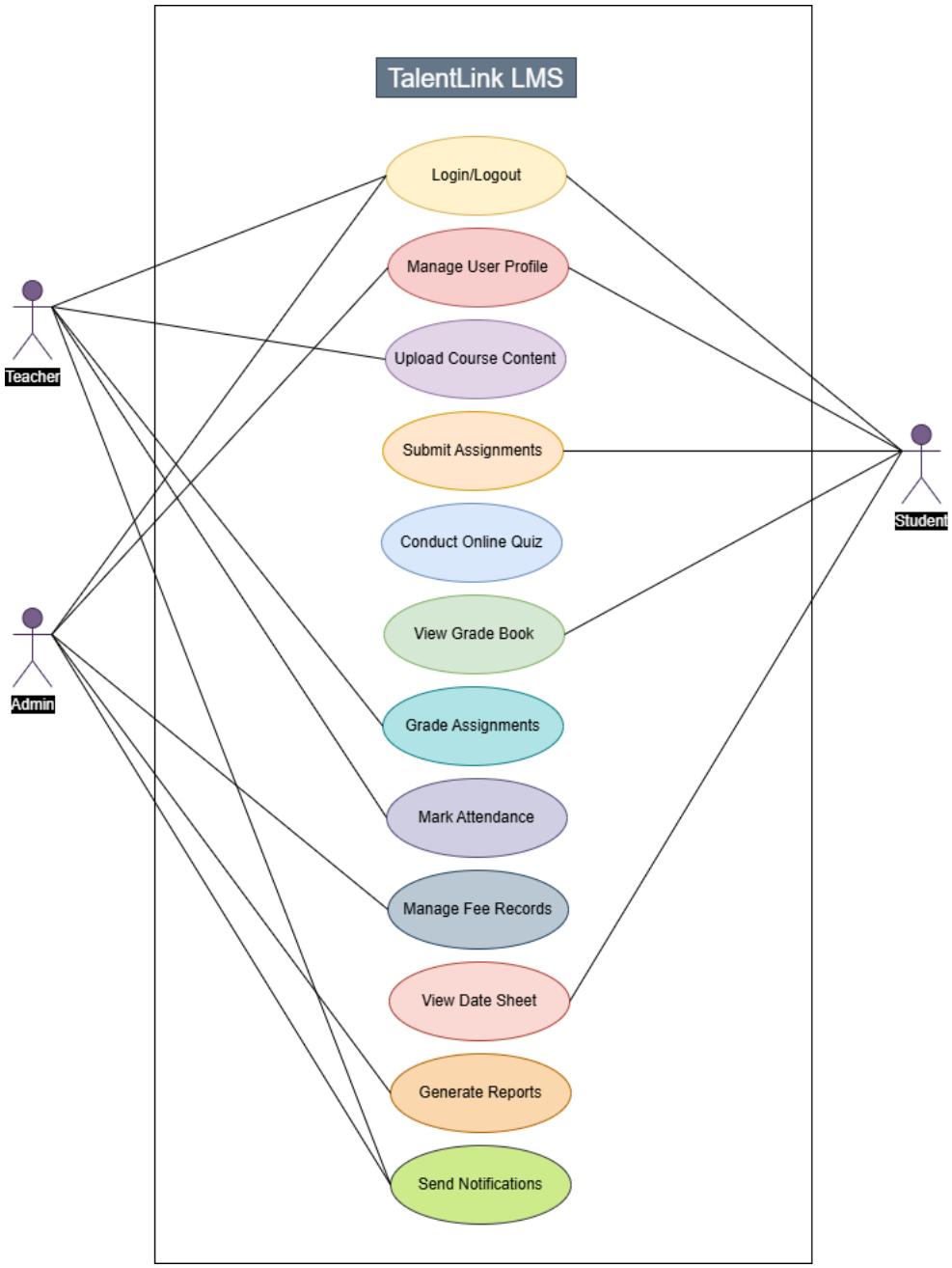
The Use Case Diagram represents the interactions between the three primary actors of the TalentLink LMS and the system's core functionalities.

Actors:

- **Admin:**
  Manages users, fee records, date sheets, and system settings.
- **Teacher**:
  Uploads course content, creates assignments, conducts quizzes, and grades students.
- **Student:**
  Views courses, submits assignments, attempts quizzes, and tracks performance.

Major Use Cases:

- Login / Logout
- Manage User Profiles
- Upload Course Content
- Submit Assignments
- Conduct Online Quiz
- View Grade Book
- Mark Attendance
- Manage Fee Records
- View Date Sheet
- Generate Reports
- Send Notifications
- Grade Assignments

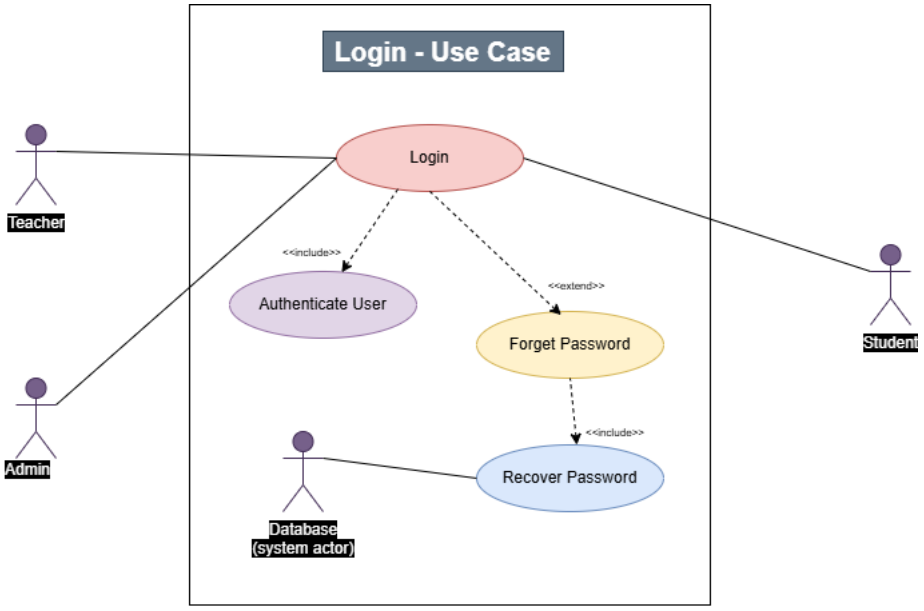

TalentLink LMS
Login/Logout
Manage User Profile
Upload Course Content
Submit Assignments
Conduct Online Quiz
View Grade Book
Grade Assignments
Mark Attendance
Manage Fee Records
View Date Sheet
Generate Reports
Send Notifications
Teacher
Admin
Student

4.2 Login Diagrams

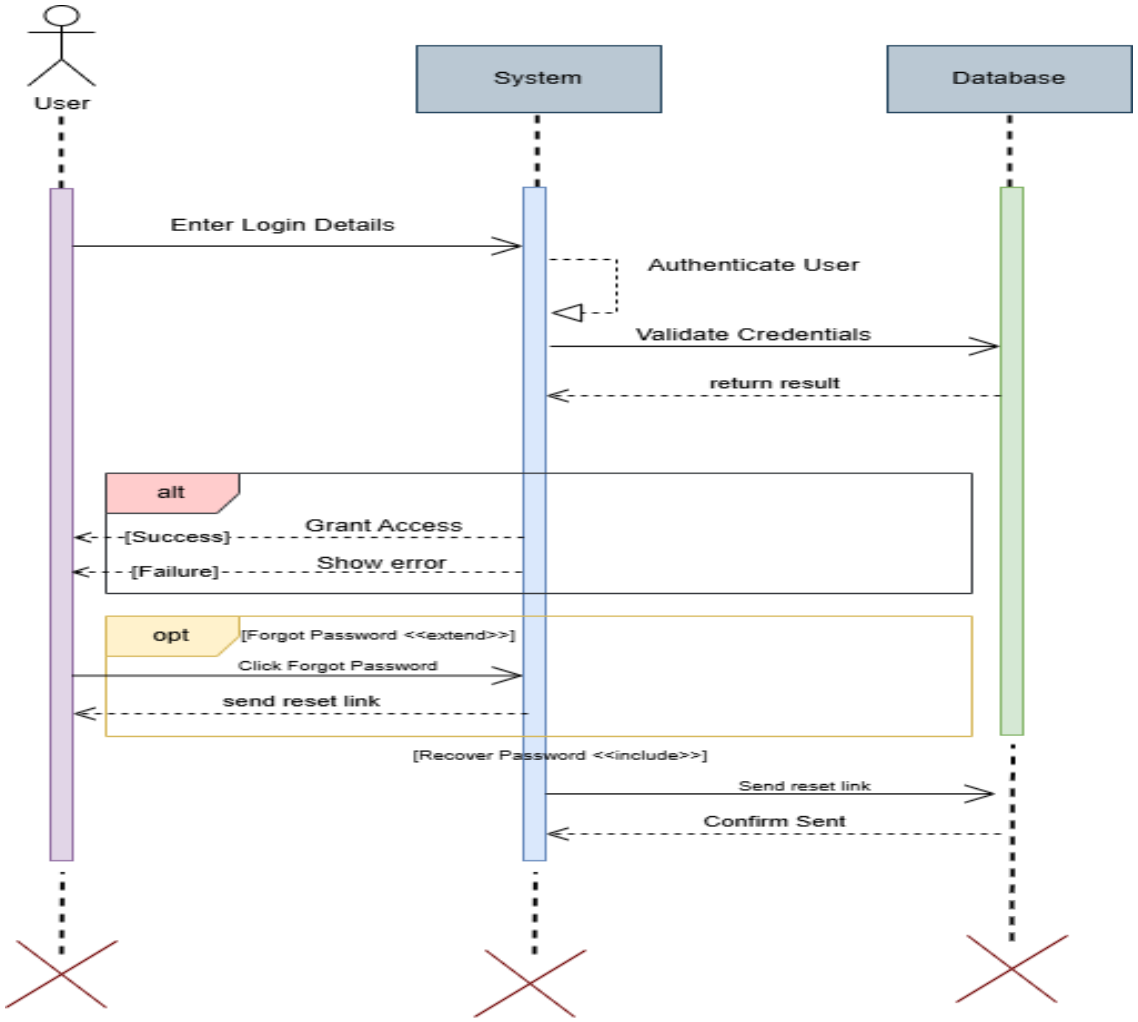
4.2.1 USE-CASE DIAGRAM FOR LOGIN

- **Description:**
  This Use Case Diagram illustrates the detailed authentication process for the TalentLink LMS. It defines how the three primary actors access the system and the underlying logic for credential verification and password recovery.
- **Actors:**
  - **Teacher:** Accesses the system to manage academic content and student evaluations.
  - **Student:** Accesses the system to interact with learning materials and track progress.
  - **Admin:** Accesses the system to manage administrative tasks and user accounts.
  - **Database (System Actor):** Performs back-end verification of user credentials and stores recovery information

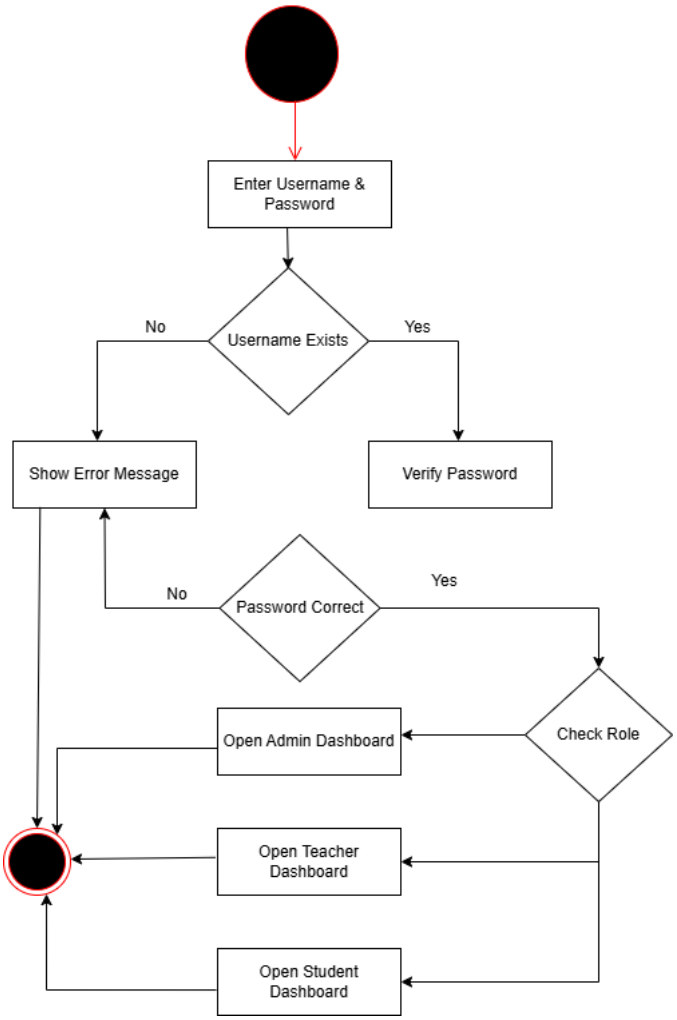
- **Major Use Cases:**
  - **Login:** The entry point for all users to gain access to the LMS.
  - **Authenticate User** (**`<<include>>`**): A mandatory internal process where the system verifies the entered credentials against the Database.
  - **Forget Password** (**`<<extend>>`**): An optional flow triggered when a user is unable to provide the correct password.
  - **Recover Password** (**`<<include>>`**): A systematic process to reset or retrieve account access, involving Database validation.



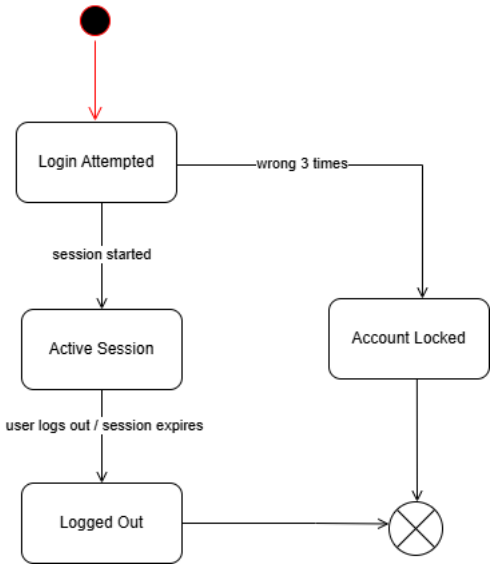
4.2.2 Login Sequence Diagram



4.2.3 Login Activity Diagram



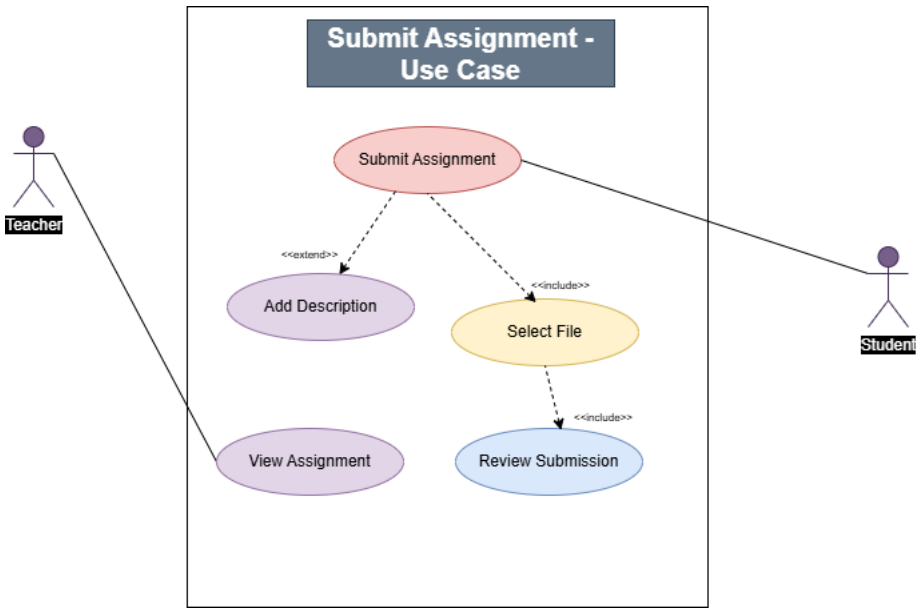
4.2.4 User Account State Diagram



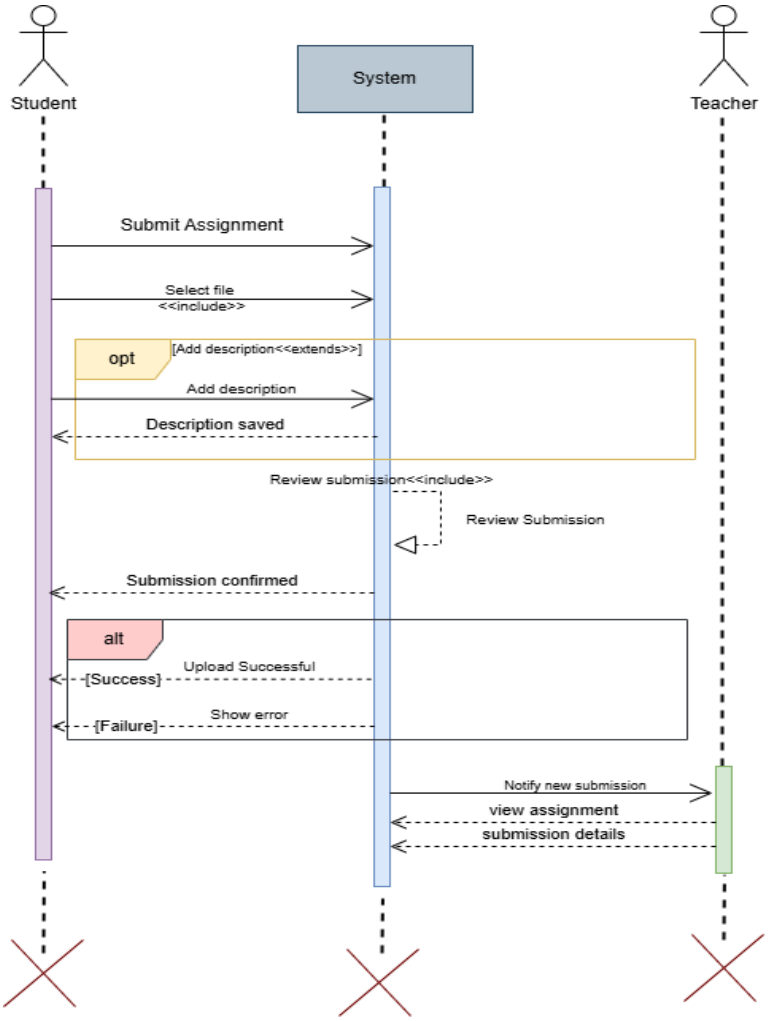
4.3 Assignment Submission Diagrams

4.3.1 Submit Assignment – Use Case

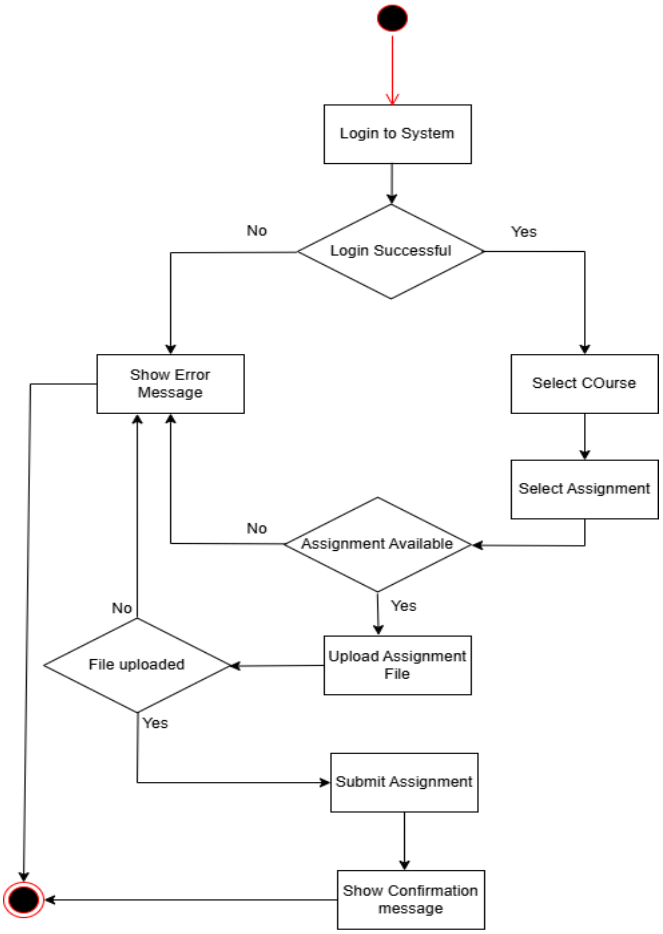
- **Description:**
  This use case details how a Student provides their academic work to the system. It ensures that the file is correctly uploaded and associated with the right course.
- **Actors:**
  - **Student (Primary):** Initiates the submission process.
  - **Teacher (Secondary):** Views the submitted assignment for evaluation.
- **Major Use Cases:**
  - **Submit Assignment:** The primary action performed by the student.
  - **Select File (<<include>>):** The mandatory step of picking a file from the device.
  - **Add Description (<<extend>>):** An optional step where the student can add comments or notes.
  - **Review Submission (<<include>>):** A final check by the system to confirm the file is uploaded before saving.
  - **View Assignment:** An action performed by the Teacher to access the student's work.



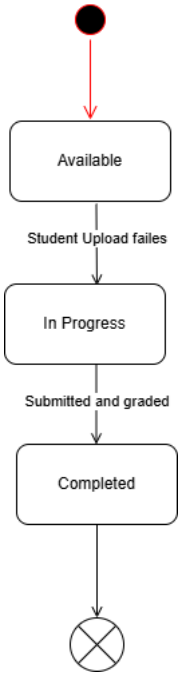
4.3.2 Submit Assignment – Sequence diagram



4.3.3 Submit Assignment_Activity Diagram



4.3.4 Submit Assignment State Diagram

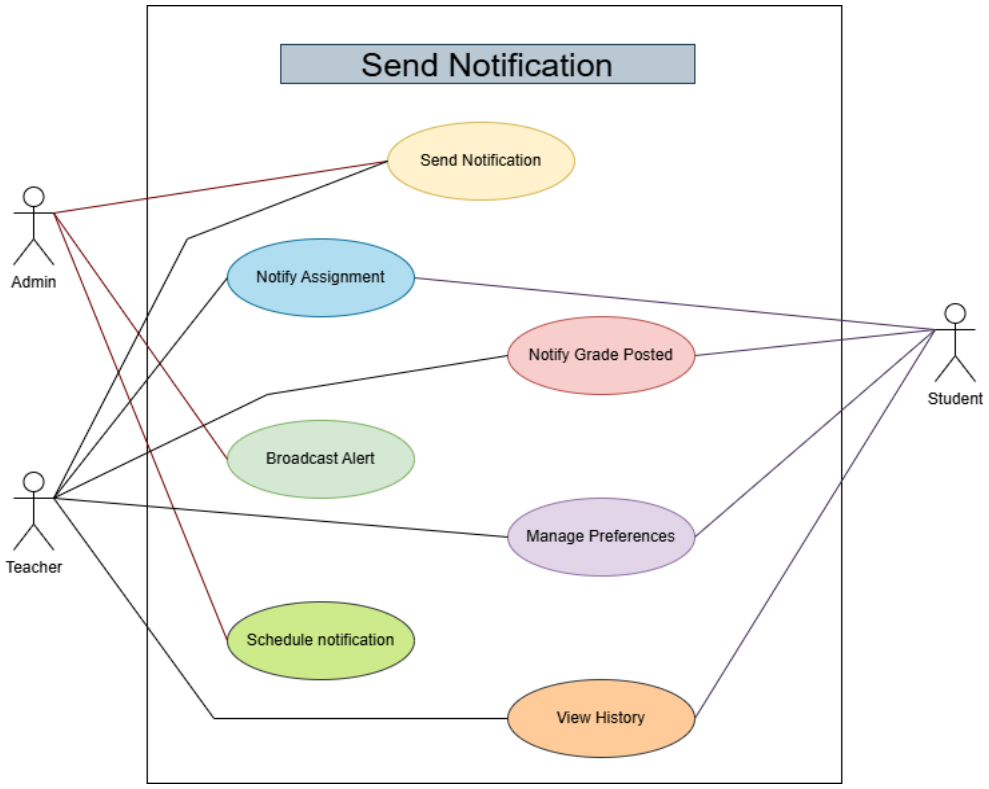


4.4 Send Notification Diagrams

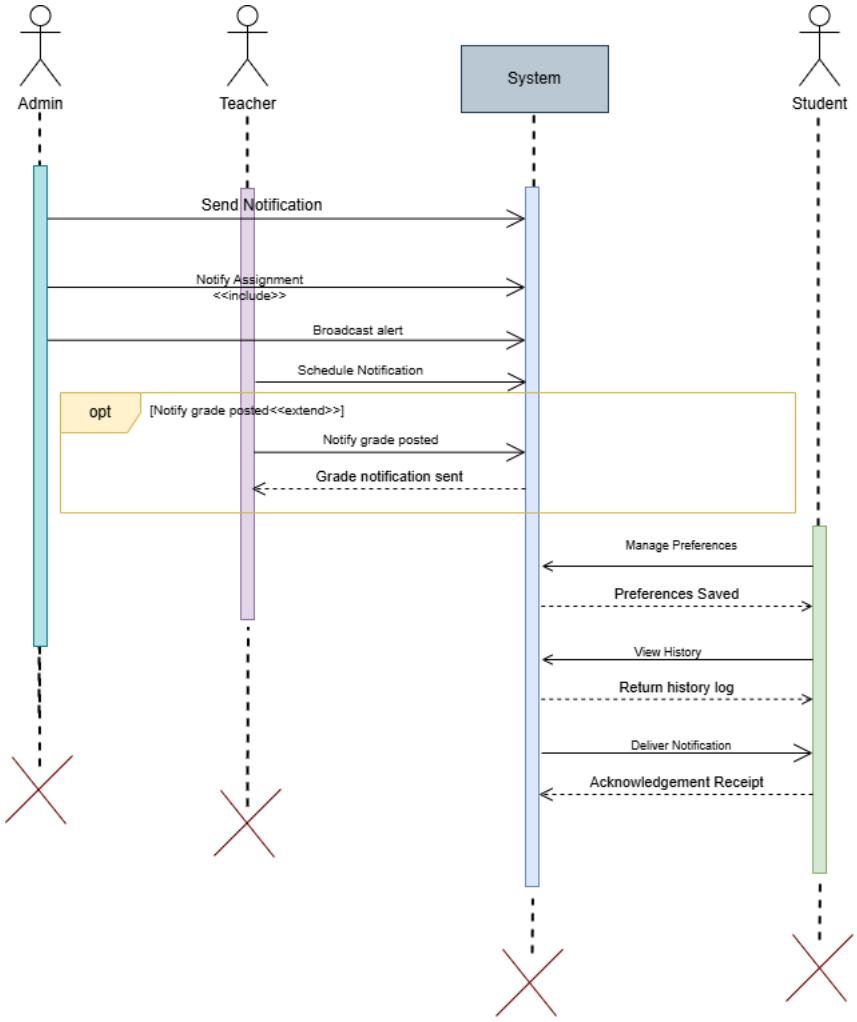
4.4.1 Use Case – Send Notification

- **Description:** This use case details how an Admin or Teacher dispatches critical updates, alerts, and academic announcements to the users within the system. It ensures that notifications are correctly generated, scheduled, or broadcasted to the intended recipients.
- **Actors:**
  - **Admin (Primary):** Initiates system-wide broadcasts and schedules notifications.
  - **Teacher (Primary):** Initiates course-specific notifications regarding assignments and grades.
  - **Student (Secondary):** Receives and views the incoming notifications.

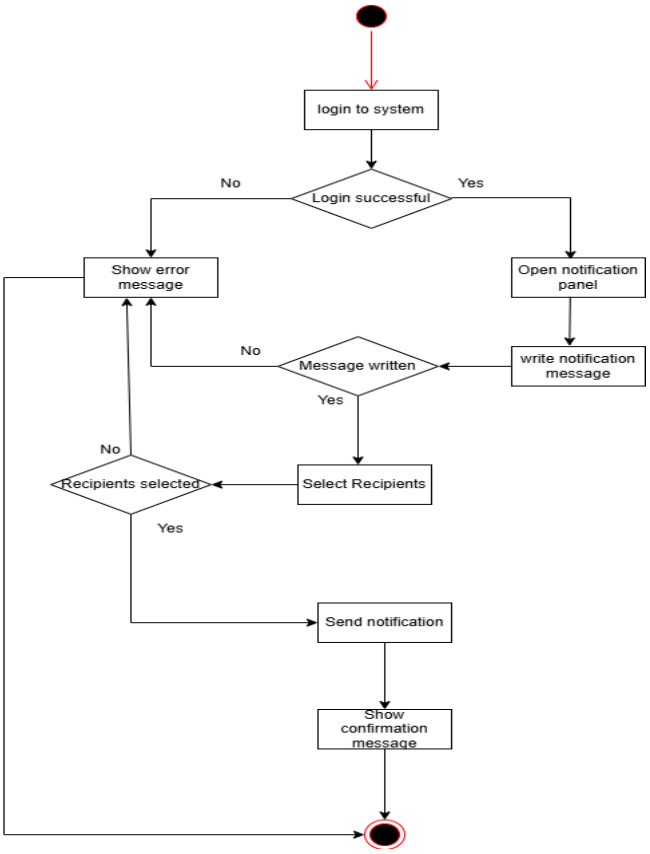
- **Major Use Cases:**
  - **Send Notification:** The primary action performed by the Admin or Teacher to compose and dispatch an alert.
  - **Select Recipients (<<include>>):** The mandatory step of specifying which user group (e.g., all students, a specific class, or an individual) will receive the notification.
  - **Schedule Time (<<extend>>):** An optional step where the Admin can set a future date and time for the notification to be sent automatically.
  - **View History:** An action performed by both Teachers and Students to access previously sent or received alerts.
  - **Manage Preferences:** An action performed by users to customize how they want to receive notifications (e.g., email or portal pop-up).


Send Notification
Send Notification
Notify Assignment
Notify Grade Posted
Broadcast Alert
Manage Preferences
Schedule notification
View History
Admin
Teacher
Student

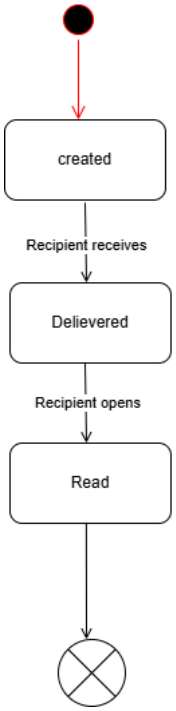
4.4.2 Send Notifications – Sequence Diagram



4.4.3 Send Notifications – Activity Diagram



4.4.4 Notification State Diagram



4.5 Data Flow Diagrams

4.5.1 Level 0 _ Context Diagram DFD

- **Description:**

  The Level 0 Context Diagram represents the entire TalentLink LMS as a single process. It provides a high-level overview of the system by showing all external actors and the data that flows between them and the system.

- **Actors:**

  Admin: Interacts with the system to manage user accounts, fee records, date sheets, and overall system configuration.

  Teacher: Interacts with the system to upload course content, create assignments, conduct quizzes, mark attendance, and submit grades.

  Student: Interacts with the system to login, access course materials, submit assignments, attempt quizzes, and view grades.

Major Data Flows:

Admin → System: User management, fee records, date sheet, system configuration.

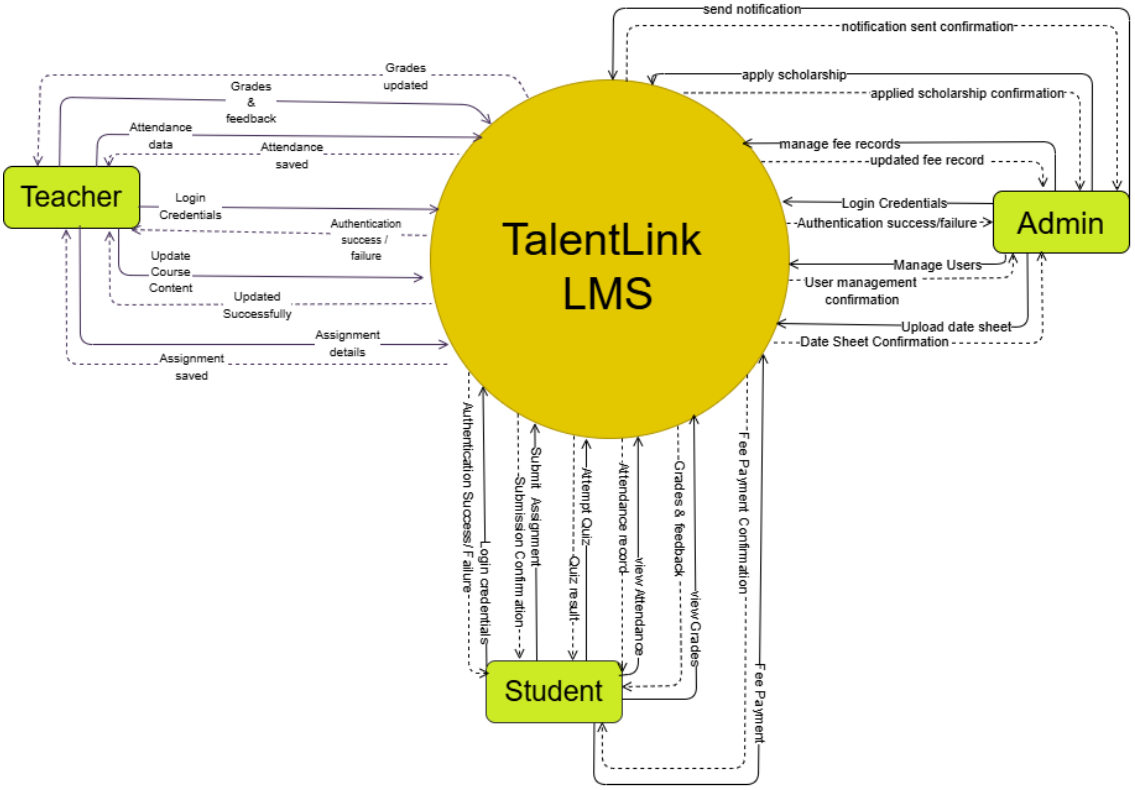
System → Admin: Reports and system alerts.

Teacher → System: Course content, assignments, quiz data, attendance, grades.

System → Teacher: Student submissions and performance reports.

Student → System: Login credentials, assignment submissions, quiz attempts, fee payment.

System → Student: Grades, notifications, course content, date sheet.



4.5.2 Level 1 _ System Level DFD

- **Description:**
  The Level 1 DFD expands the TalentLink LMS into its core internal processes. It shows how data flows between external entities, internal processes, and data stores in detail.
- **Processes:**

User Authentication:
Handles login credentials and provides dashboard access to Admin, Teacher, and Student.

Course Management:
Manages course content uploaded by Teacher and stores it in Course DB.

Assignment Management:
Handles assignment submission by Student and assignment details by Teacher, stores in Assignment DB.

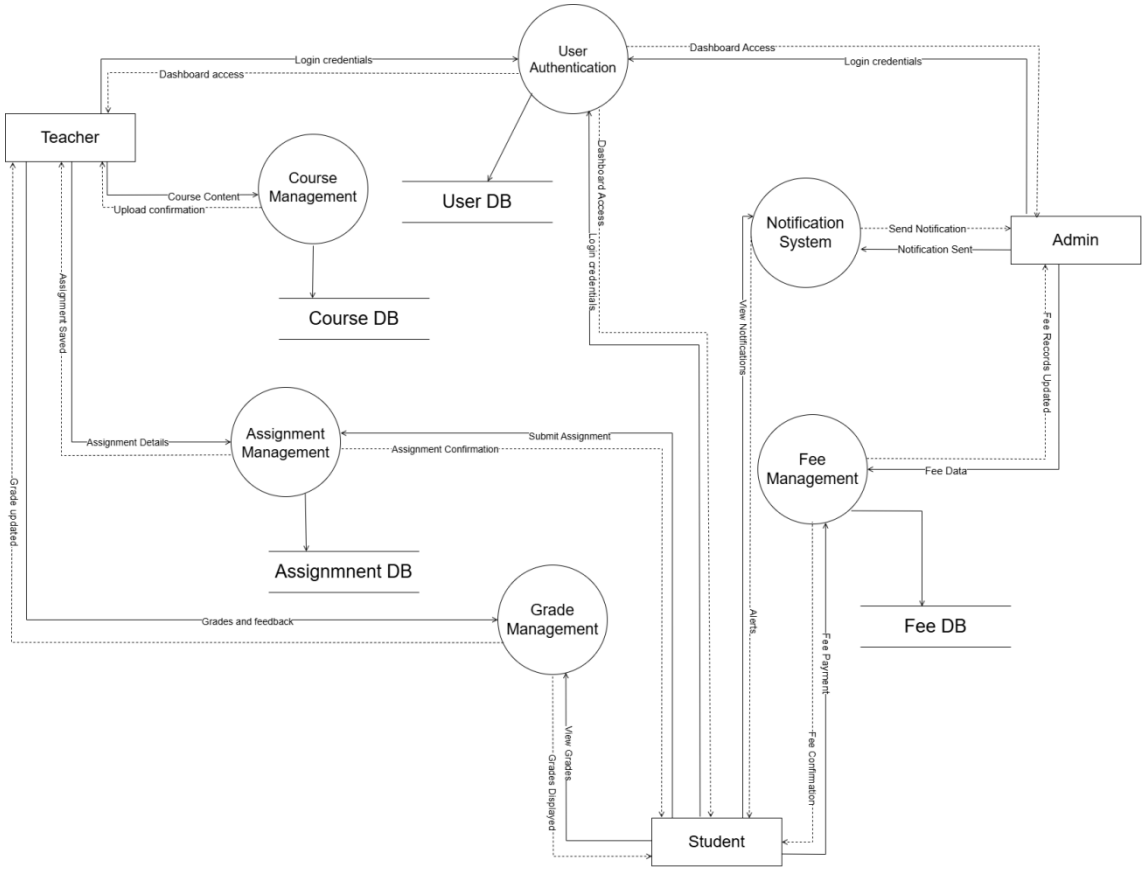
Grade Management:
Manages grades and feedback submitted by Teacher and displays results to Student.

Fee Management:
Handles fee data from Admin and fee payment from Student, stores in Fee DB.

Notification System:
Manages notifications sent by Admin and delivers alerts to Teacher and Student.

Data Stores:
User DB: Stores user login and authentication records.
Course DB: Stores all course content and materials.

Assignment DB: Stores assignment details and student submissions.
Fee DB: Stores fee records and payment history.



4.5.3 Level 3 _Detailed Level DFD

4.5.3.1 User Authentication

Description:

The Level 2 DFD for User Authentication expands the User Authentication process from Level 1. It shows the detailed steps involved in verifying the identity of Admin, Teacher, and Student before granting access to the system.

Sub-Processes:

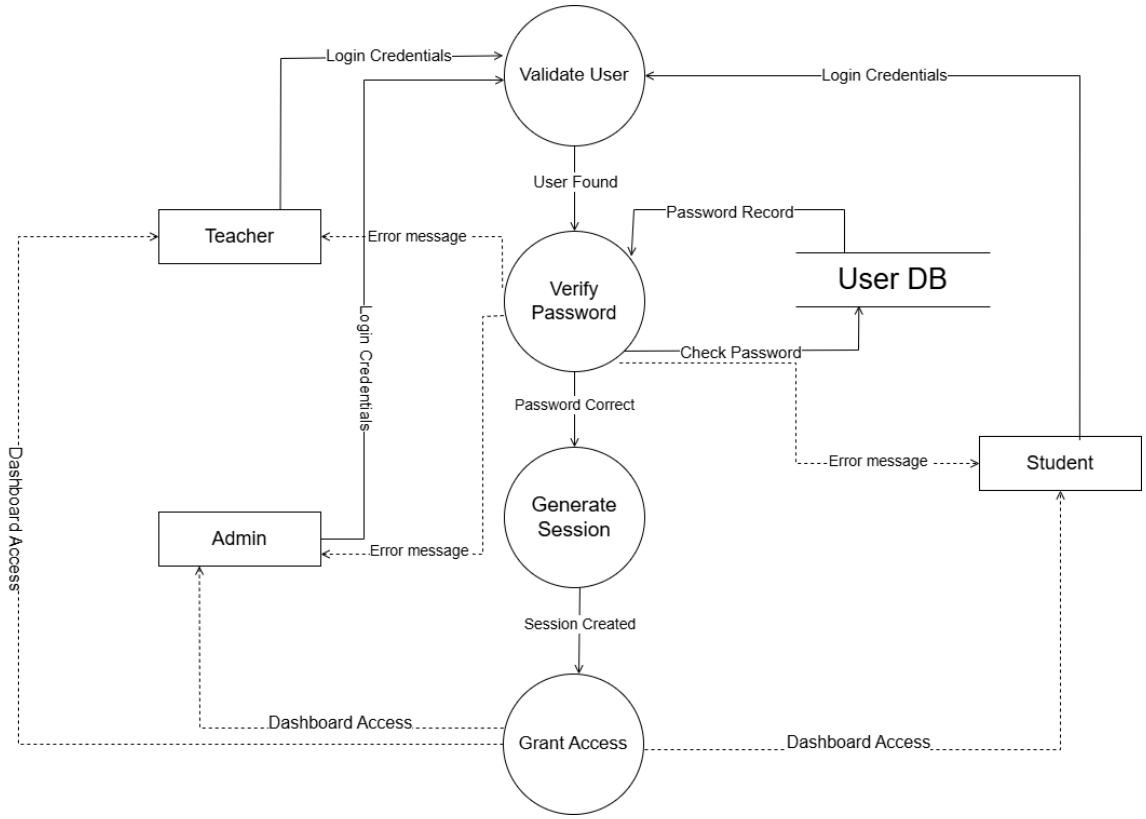
- **Validate User:** Checks whether the entered username exists in the User DB.
- **Verify Password:** Matches the entered password with the stored encrypted password.
- **Generate Session:** Creates a session token upon successful authentication.
- **Grant Access:** Redirects the user to their respective dashboard based on their role.

Entities:

- **Admin, Teacher, Student:** All three enter login credentials to access the system.
- **Data Store:**
- **User DB:** Contains all user records including username, password, and role.

- **Data Flows:**

User enters login credentials → Validate User
Validate User → User DB (check username)
User DB → Verify Password (return record)
Verify Password → Generate Session (if correct)
Generate Session → Grant Access
Grant Access → User (Dashboard access)
Verify Password → User (Error message if wrong)



4.6 Class Diagram

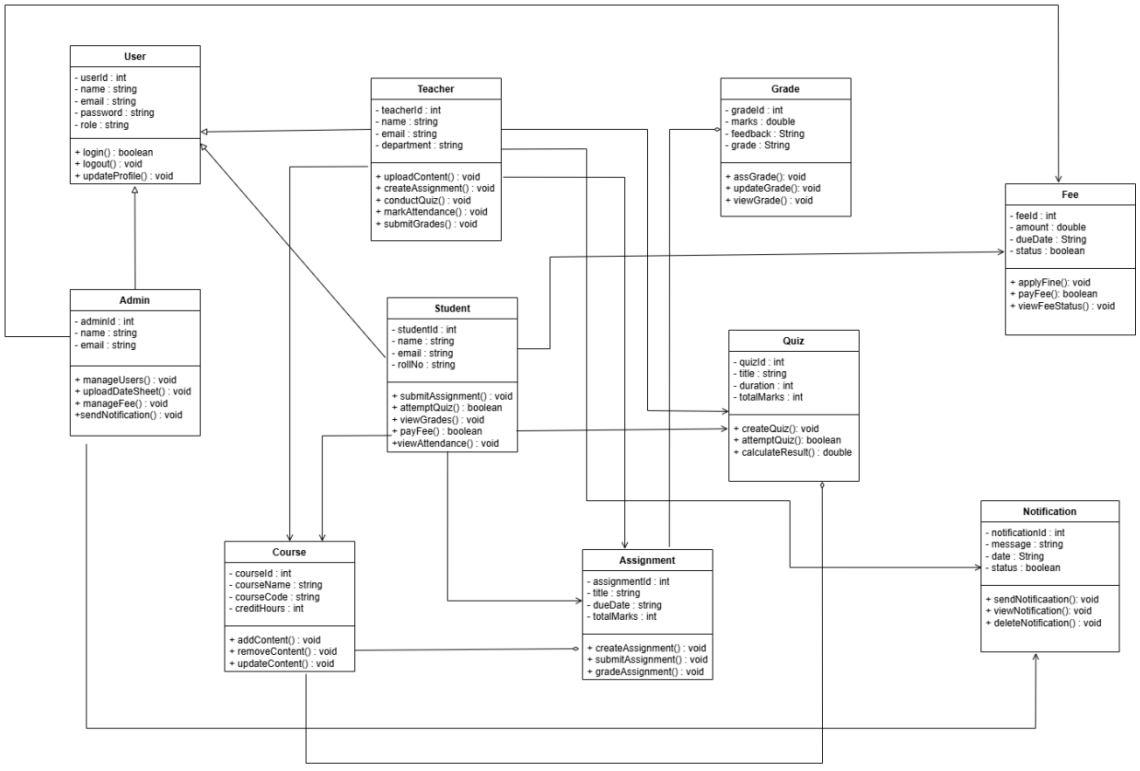
A Class Diagram represents the static structure of the TalentLink LMS. It shows the system's classes, their attributes, methods, and the relationships between them.

Classes:

- **User:** Base class for Admin, Teacher, and Student containing common attributes like userID, name, email, and password.
- **Admin:** Manages users, fee records, date sheets, and notifications.
- **Teacher:** Uploads course content, creates assignments, conducts quizzes, marks attendance, and submits grades.
- **Student:** Submits assignments, attempts quizzes, views grades, pays fee, and views attendance.
- **Course:** Contains course content, assignments, and quizzes.
- **Assignment:** Manages assignment creation, submission, and grading.
- **Quiz:** Handles quiz creation, attempts, and result calculation.
- **Grade:** Stores marks, feedback, and grade of each student.
- **Fee:** Manages fee payment, fines, and fee status.
- **Notification:** Handles sending and receiving of notifications.

Relationships:

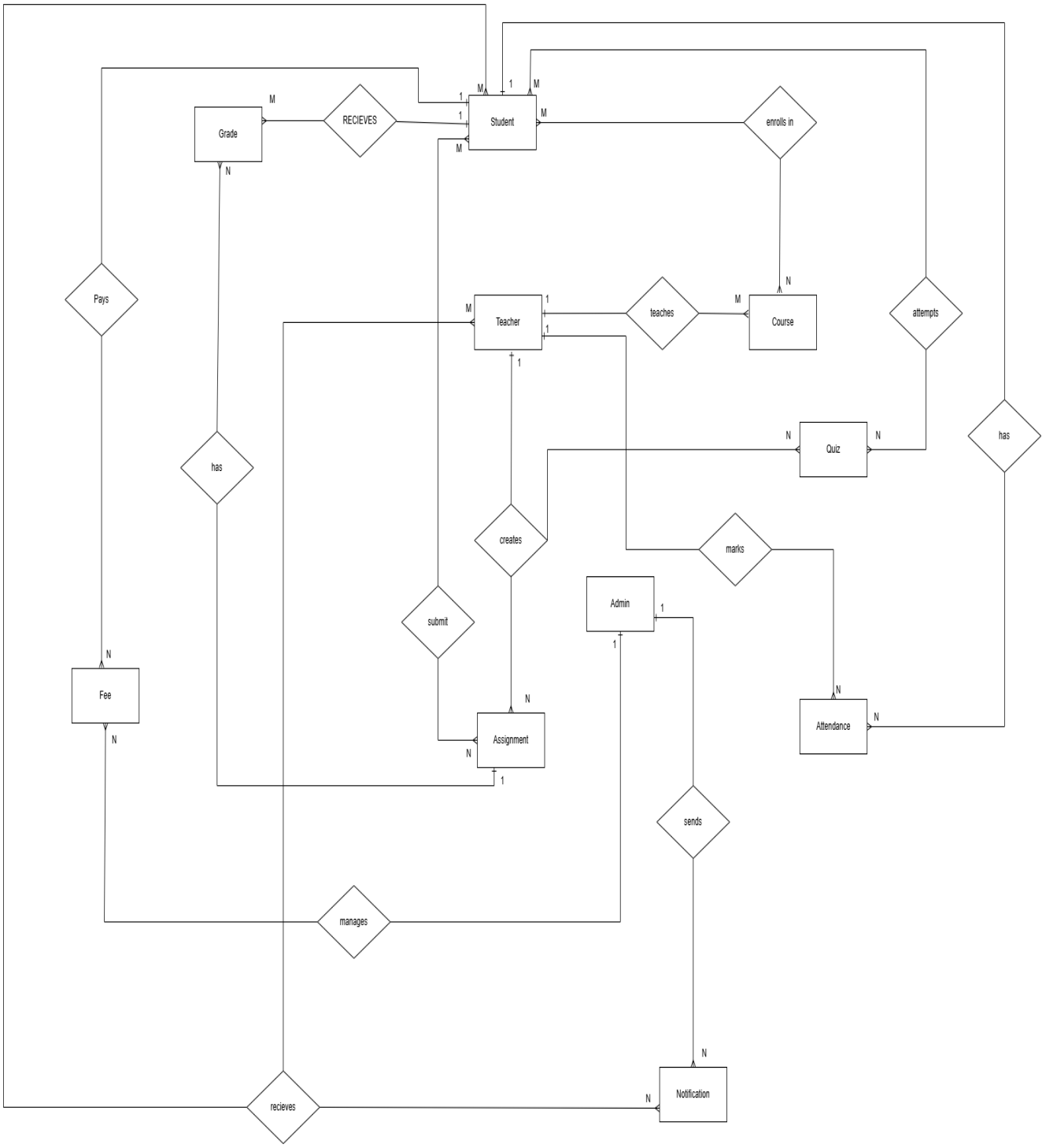
- **Inheritance:** Admin, Teacher, and Student inherit from User class.
- **Association:** Teacher and Student are associated with Course, Assignment, and Quiz.
- **Aggregation:** Assignment and Quiz are part of Course. Grade is part of Assignment.



4.7 ERD

Description:

The Entity Relationship Diagram (ERD) of TalentLink LMS represents the database structure of the system. It shows all the entities, their relationships, and the cardinality between them.



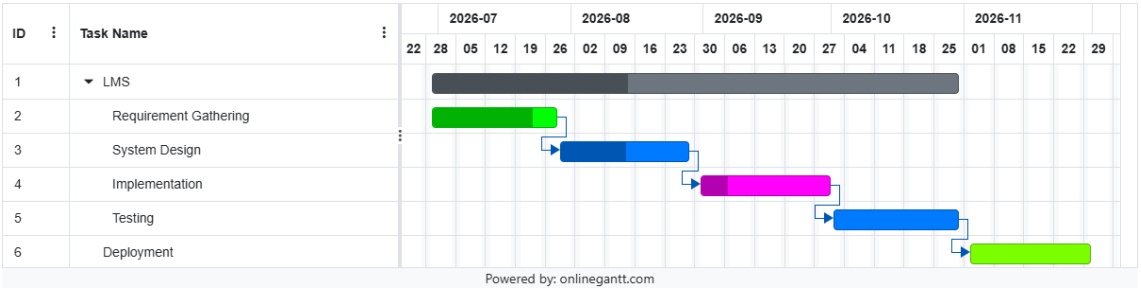
4.8 Gantt Chart

Description:

The Gantt Chart represents the project timeline for the development of TalentLink LMS based on the Waterfall Model. It shows the sequence of phases, their start dates, and durations over a period of 6 months.

Phases:

- **Requirement Gathering:** The first phase where all system requirements are gathered and documented. *(July 2026)*
- **System Design:** The second phase where system architecture, diagrams, and database design are created. *(August 2026)*
- **Implementation:** The third phase where frontend and backend development takes place. *(September 2026)*
- **Testing:** The fourth phase where the system is tested for bugs and errors. *(October 2026)*
- **Deployment:** The final phase where the system is launched and delivered to the client. *(November 2026)*



4.9 Test Cases:

4.9.1 User Login Test Case

Positive

Test Case ID : TC-LG-01	**Test Designed by:** [Your Name]
Test Priority: High	**Test Designed date:** 29-06-2026
Module Name: User Authentication	**Test Executed by:** [Your Name]
Test Title: Verify that user can login successfully with valid credentials	**Test Execution date:** 29-06-2026

Description: Test the login functionality with valid username and password.
Pre-conditions: User must be registered in the system.
Dependencies: User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Login Page	Browser URL	Login Page displayed	Login page displayed	Pass	-
2	Enter Username	tayyab@lms.com	Username accepted	Username accepted	Pass	-
3	Enter Password	Tayyab.123	Password accepted	Password accepted	Pass	-
4	Click Login	Login Command	User redirected to dashboard	User redirected to dashboard	Pass	-

Post-conditions: User session created. Dashboard displayed successfully.

Negative

Test Case ID : TC-LG-02	**Test Designed by:** [Your Name]
Test Priority: High	**Test Designed date:** 29-06-2026
Module Name: User Authentication	**Test Executed by:** [Your Name]
Test Title: Verify that user cannot login with invalid credentials	**Test Execution date:** 29-06-2026

Description: Test the login functionality with invalid username and password.
Pre-conditions: User enters wrong credentials.
Dependencies: User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Login Page	Browser URL	Login Page displayed	Login page displayed	Pass	-
2	Enter	tayyab@lms.com	Username	Username	Pass	-

	Username		accepted	accepted		
3	Enter wrong Password	Tayyab..123	Password Field accepted	Password field accepted	Pass	-
4	Click Login	Login Command	Error message displayed "Invalid username or password"	Error message displayed	Fail	Invalid credentials

Post-conditions: User not logged in. Error message displayed on screen.

4.9.2 Submit Assignment Test Case

Positive

Test Case ID : TC-SA-01	**Test Designed by:** [Your Name]
Test Priority: High	**Test Designed date:** 29-06-2026
Module Name: Assignment Management	**Test Executed by:** [Your Name]
Test Title: Verify that student can submit assignment successfully	**Test Execution date:** 29-06-2026

Description: Test the assignment submission functionality with valid file upload.
Pre-conditions: Student must be logged in. Assignment must be available.
Dependencies: Assignment Database, User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Assignment Page	Student Dashboard	Assignment page displayed	Assignment page displayed	Pass	-
2	Select Course	Course Name	Course Selected Successfully	Course Selected	Pass	-
3	Select Assignment	Assignment Title	Assignment opened successfully	Assignment opened	Pass	-
4	Upload File	Submit Command	Confirmation message displayed	Confirmation displayed	Pass	-
5	Click Submit Button	Submit Command	Confirmation message displayed	Confirmation displayed	Pass	-

Post-conditions: Assignment saved in database. Confirmation message displayed.

Negative

Test Case ID : TC-SA-02	**Test Designed by:** [Your Name]
Test Priority: High	**Test Designed date:** 29-06-2026
Module Name: Assignment Management	**Test Executed by:** [Your Name]
Test Title: Verify that student cannot submit assignment uploading file	**Test Execution date:** 29-06-2026

Description: Test the assignment submission functionality without uploading any file.
Pre-conditions: Student must be logged in. Assignment must be available.
Dependencies: Assignment Database, User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Assignment Page	Student Dashboard	Assignment page displayed	Assignment page displayed	Pass	-
2	Select Course	Course Name	Course Selected Successfully	Course Selected	Pass	-
3	Select Assignment	Assignment Title	Assignment opened successfully	Assignment opened	Pass	-
4	Skip upload file	No File	Error message displayed	Error message displayed	Fail	No file uploaded
5	Click Submit Button	Submit Command	Submission blocked	Submission blocked	Fail	No file uploaded

Post-conditions: Assignment not submitted. Error message displayed on screen.

4.9.3 Send Notification Test Case

Positive

Test Case ID : TC-SN-01	**Test Designed by:** [Your Name]
Test Priority: Medium	**Test Designed date:** 29-06-2026
Module Name: Notification System	**Test Executed by:** [Your Name]
Test Title: Verify that Admin can send notification successfully	**Test Execution date:** 29-06-2026

Description: Test the notification functionality with valid message and recipients.
Pre-conditions: Admin must be logged in. Recipients must be registered in system.
Dependencies: Notification Database, User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Notification Panel	Admin Dashboard	Notification panel displayed	Notification panel displayed	Pass	-
2	Write message	“Exam scheduled	Message written	Message written	Pass	-

		tomorrow”	successfully			
3	Select Recipients	All Students	Recipients selected successfully	Recipients selected	Pass	-
4	Click Send Button	Send Command	Notification sent successfully	Notification sent	Pass	-
5	Check Confirmation	System Response	Confirmation message displayed	Confirmation displayed	Pass	-

Post-conditions: Notification saved in database. Recipients notified successfully.

Negative

Test Case ID : TC-SN-02	**Test Designed by:** [Your Name]
Test Priority: Medium	**Test Designed date:** 29-06-2026
Module Name: Notification System	**Test Executed by:** [Your Name]
Test Title: Verify that Admin cannot send notification without writing message or selecting recipients	**Test Execution date:** 29-06-2026

Description: Test the notification functionality without writing message and without selecting recipients.
Pre-conditions: Admin must be logged in.
Dependencies: Notification Database, User Database

Step	Test Step	Test Data	Expected Result	Actual Result	Status	Notes
1	Open Notification Panel	Admin Dashboard	Notification panel displayed	Notification panel displayed	Pass	-
2	Skip message	No message	Error message displayed	Error message displayed	Fail	Message Required
3	Skip Recipients	No Recipients	Error message displayed	Error message displayed	Fail	Recipients Required
4	Click Send Button	Send Command	Notification blocked	Notification blocked	Fail	Fields Required

Post-conditions: Notification not sent. Error message displayed on screen.